

# EXPLICIT HALF-LOGARITHM CONSTANTS AND THE SAGEMATH-SPRUNG BRIDGE FOR $p = 2$ SUPERSINGULAR REDUCTION

RAPHAEL ELKUCH

**ABSTRACT.** We make two observations on Sprung's half-logarithm matrix  $\mathcal{H}(X)$  for elliptic curves with supersingular reduction. First, for any prime  $p$ , the projective-limit construction of  $\mathcal{H}(X)$  at  $X = 0$  degenerates to the closed form  $A^{-2}B$ , since the cyclotomic-polynomial values  $\Phi_{p^k}(1)$  collapse to  $p$ . Second, by reading the SageMath source code for `padic_lseries.Dp_valued_series`, we recover the explicit identification between Sage's output and the constants of  $L_p(E, \alpha, T)$  in the  $(1, \alpha)$ -basis: Sage returns  $(1 - \varphi)^{-2} \cdot L_p$  in the Dieudonné  $(\omega, \varphi\omega)$ -basis, with  $\varphi$  the half-frobenius matrix at the supersingular prime. Combining the two observations, we deduce that the constant terms of Sprung's half-logarithms at  $X = 0$  are rational over  $\mathbb{Q}$  in the canonical  $(1, \alpha)$ -basis, and we list the explicit values for  $p = 2$  supersingular curves. We verify the SageMath–Sprung bridge bit-exactly on 180 elliptic curves drawn from the Cremona database with conductors in [37, 63 096].

## 1. INTRODUCTION

Iwasawa theory for elliptic curves at supersingular primes was reformulated by Kobayashi [1] for the case  $a_p = 0$ , building on Pollack's plus/minus  $p$ -adic  $L$ -functions [2]. Sprung [3] extended the construction to all supersingular reductions with  $a_p \neq 0$ , introducing a pair of half-logarithms  $\log_\alpha^\sharp$ ,  $\log_\alpha^b$  and a pair of  $p$ -adic  $L$ -functions  $L_p^\sharp$ ,  $L_p^b$  satisfying the canonical decomposition ([3, Theorem 6.12])

$$(1) \quad L_p(E, \alpha, X) = \log_\alpha^\sharp(1 + X) \cdot L_p^\sharp(E, X) + \log_\alpha^b(1 + X) \cdot L_p^b(E, X).$$

The half-logarithm matrix

$$(2) \quad \begin{pmatrix} \log_\alpha^\sharp(1 + X) & \log_{\bar{\alpha}}^\sharp(1 + X) \\ \log_\alpha^b(1 + X) & \log_{\bar{\alpha}}^b(1 + X) \end{pmatrix} := \mathcal{H}(X) \cdot \begin{pmatrix} 1 & 1 \\ -\alpha^{-1} & -\bar{\alpha}^{-1} \end{pmatrix}$$

is constructed ([3, Definition 6.8]) from the auxiliary matrix

$$(3) \quad \mathcal{H}(X) := \lim_{n \rightarrow \infty} C_1(X) \cdot C_2(X) \cdots C_n(X) \cdot A^{-(n+2)}$$

---

*Date:* May 26, 2026.

*2020 Mathematics Subject Classification.* Primary 11G05, 11G40, 11R23; Secondary 11Y40.

*Key words and phrases.*  $p$ -adic  $L$ -function, supersingular reduction, Iwasawa theory, half-logarithm, Sprung's main conjecture, SageMath,  $p = 2$  elliptic curves, Dieudonné module.

where  $C_k(X) = \begin{pmatrix} a_p & 1 \\ -\Phi_{p^k}(1+X) & 0 \end{pmatrix}$ ,  $A = \begin{pmatrix} a_p & 1 \\ -p & 0 \end{pmatrix}$  is the Frobenius action matrix at  $X = 0$ , and  $\alpha, \bar{\alpha}$  are the roots of  $X^2 - a_p X + p$ . Equations (2) and (3) follow Sprung's Definitions 6.8 and 4.5 directly; no modification of his framework is required.

The case  $p = 2$  is delicate: Sprung's main theorem in [4] is stated for *odd* supersingular primes, leaving the  $p = 2$  case open in the published literature, although Lei [6] remarks that his two-variable factorisation theorem extends to  $p = 2$  with minor notational adjustments. Numerical investigation of the construction at  $p = 2$  requires interfacing with computer algebra. The SageMath system [7] provides the routine `padic_lseries.Dp_valued_series`, which returns a power series in  $T$  with values in the Dieudonné module  $D_p(E) = H_{\text{dR}}^1(E/\mathbb{Q}_p)$ , but the precise interpretation of its output requires reading the source.

In this note we make two observations.

(I) *Triviality of the limit at  $X = 0$ .* For any prime  $p$ , the projective limit (3) collapses at  $X = 0$ : since  $\Phi_{p^k}(1) = p$  for all  $k \geq 1$ , the cofactor matrices  $C_k(0)$  coincide with  $A$ , and  $C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^{-2}$  holds independently of  $n$ . This is Theorem 3.1 below.

(II) *The SageMath–Sprung bridge.* Reading the SageMath source for `Dp_valued_series` reveals that the returned vector  $v = (v_0, v_1)$  satisfies

$$(1 - \varphi)^{-2} \cdot L_p(E, T) = v_0 \cdot \omega + v_1 \cdot \varphi(\omega)$$

in the Dieudonné basis  $(\omega, \varphi\omega)$ , where  $\varphi$  is the Frobenius matrix represented by  $\begin{pmatrix} 0 & -1/p \\ 1 & a_p/p \end{pmatrix}$  in this basis. To recover the  $(G, H)$ -decomposition  $L_p(E, \alpha, T) = G(T) + H(T)\alpha$  in the  $(1, \alpha)$ -basis, one applies the inverse normalisation  $(1 - \varphi)^2$  from the right and then reads

$$H = -\frac{1}{p}(\text{second component}), \quad G = (\text{first component}) - a_p H.$$

This is Theorem 3.6.

Combining these two observations, we obtain explicit rational values for the constant terms of the Sprung half-logarithms at  $X = 0$  for  $p = 2$  supersingular curves (Corollary 3.5). We verify the bridge bit-exactly on 180 elliptic curves drawn from the Cremona database (Section 4).

**Acknowledgments.** The author thanks the SageMath developer community and J. E. Cremona for the elliptic-curve infrastructure on which this work depends. Manuscript preparation and bulk verification benefited from interactive work with Anthropic's Claude; all mathematical statements were independently verified by the author.

## 2. SETTING AND PRELIMINARIES

**2.1. Notation.** Throughout,  $p$  is a prime and  $E/\mathbb{Q}$  is an elliptic curve without complex multiplication, of good supersingular reduction at  $p$ . Let  $a_p = a_p(E)$ , with  $|a_p| < 2\sqrt{p}$  and  $p \mid a_p$  by supersingularity. At  $p = 2$  this gives  $a_p \in \{-2, 0, +2\}$ ; we treat all three cases in Section 3. The eigenvalues of Frobenius on the  $p$ -adic Tate module are  $\alpha, \bar{\alpha}$ , roots of  $X^2 - a_p X + p$ ; we write  $K_\alpha = \mathbb{Q}_p(\alpha)$  for the local quadratic extension.

**2.2. The Sprung half-logarithm matrix.** The auxiliary matrices in (3) are

$$(4) \quad A = \begin{pmatrix} a_p & 1 \\ -p & 0 \end{pmatrix}, \quad C_k(X) = \begin{pmatrix} a_p & 1 \\ -\Phi_{p^k}(1+X) & 0 \end{pmatrix}.$$

The cyclotomic polynomial values  $\Phi_{p^k}(1)$  collapse to a single value:

**Lemma 2.1.** *For every prime  $p$  and every  $k \geq 1$ , we have  $\Phi_{p^k}(1) = p$ .*

*Proof.* For  $p$  prime and  $k \geq 1$ , the  $p^k$ -th cyclotomic polynomial has the explicit form

$$\Phi_{p^k}(X) = \sum_{j=0}^{p-1} X^{jp^{k-1}} = 1 + X^{p^{k-1}} + X^{2p^{k-1}} + \dots + X^{(p-1)p^{k-1}},$$

a sum of  $p$  terms. Each term evaluates to 1 at  $X = 1$ , so  $\Phi_{p^k}(1) = p$ .  $\square$

**2.3. The Dieudonné basis and the Frobenius matrix.** On the Dieudonné module  $D_p(E) = H_{\text{dR}}^1(E/\mathbb{Q}_p)$ , the Frobenius operator  $\varphi$  admits the matrix representation

$$(5) \quad \varphi = \begin{pmatrix} 0 & -1/p \\ 1 & a_p/p \end{pmatrix}$$

in the *cyclic* basis  $(\omega, \varphi\omega)$ , where  $\omega$  is the invariant differential. This is the companion matrix of the characteristic polynomial  $X^2 - (a_p/p)X + 1/p = 0$  of  $\varphi$ , and it satisfies  $\varphi^2 - (a_p/p)\varphi + 1/p = 0$ .

**Remark 2.2** (Two distinct Frobenius representations in SageMath). SageMath uses two different bases for the Dieudonné module, leading to two different matrix representations of the same operator  $\varphi$ :

- (a) The *geometric* basis  $(\omega, \eta)$  with  $\eta = x\omega$ , returned by the method `L.frobenius()`.
- (b) The *cyclic* basis  $(\omega, \varphi\omega)$ , used internally in `L.Dp_valued_series()` via the explicit assignment `phi = matrix([[0, -1/p], [1, E.ap(p)/p]])` at `padic_lseries.py`, line 1442.

The matrix (5) represents  $\varphi$  in basis (b); this is the convention we use throughout, since the bridge formula of Theorem 3.6 concerns exactly the output of `Dp_valued_series`. A reader using `L.frobenius()` to obtain (5) would obtain a different matrix, related to ours by an explicit change-of-basis.

We will encounter the matrix  $\varepsilon^{-1} := (1 - \varphi)^2$  throughout. Direct computation gives:

$$\text{Lemma 2.3. } \det((1 - \varphi)^2) = \frac{|E(\mathbb{F}_p)|^2}{p^2} = \frac{(p + 1 - a_p)^2}{p^2}.$$

*Proof.*  $\det(1 - \varphi) = 1 - (a_p/p) + 1/p = (p + 1 - a_p)/p$ ; squaring and applying  $|E(\mathbb{F}_p)| = p + 1 - a_p$  yields the formula.  $\square$

In our setting  $p = 2$ , this becomes

$$(6) \quad \det((1 - \varphi)^2) = \frac{(3 - a_p)^2}{4} = \begin{cases} 25/4 & \text{if } a_p = -2, \\ 9/4 & \text{if } a_p = 0. \end{cases}$$

## 3. MAIN RESULTS

3.1. Triviality of the  $X = 0$  limit.

**Theorem 3.1** (Triviality of the limit at  $X = 0$ , for any prime  $p$ ). *Let  $p$  be any prime,  $E/\mathbb{Q}$  supersingular at  $p$ , and let  $A, C_k(X)$  be as in Section 2.2. The truncated product*

$$C_1(0) \cdot C_2(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^{-2}$$

*is independent of  $n \geq 1$ . In particular,*

$$(7) \quad \mathcal{H}(0) = A^{-2}.$$

*Proof.* By Lemma 2.1,  $\Phi_{p^k}(1) = p$  for every  $k \geq 1$ , so  $C_k(0) = A$  universally. Hence  $C_1(0) \cdots C_n(0) = A^n$  and

$$C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^n \cdot A^{-(n+2)} = A^{-2},$$

independent of  $n$ . Passing to the projective limit yields the formula.  $\square$

**Remark 3.2.** The statement and proof are valid for every prime  $p$ , not only  $p = 2$ . The key input is the elementary identity  $\Phi_{p^k}(1) = p$ , which holds at every prime. We will apply Theorem 3.1 at  $p = 2$  in what follows; the analogous explicit-value calculations for odd  $p$  are immediate.

**Remark 3.3** (Relation to Sprung's general convergence theorem). The convergence of the projective limit (3) for general  $X$  is the content of Sprung's [3, Lemma 4.4 (Convergence Lemma) and Lemma 4.8 (Growth Lemma)], and is reformulated by Lei [6, Theorem 1.5] in the language of  $p$ -adic logarithmic matrices. Theorem 3.1 above is the specialisation  $X = 0$ , where the limit becomes algebraically trivial. The determinant of  $\mathcal{H}(X)$  is computed in [3, Example 4.7]; for our application at  $X = 0$  it reduces to  $\det(A^{-2}) = 1/p^2$ , in agreement with Lemma 2.3 via the bridge  $\varepsilon = (1 - \varphi)^2$ .

**Proposition 3.4** (Explicit  $\mathcal{H}(0)$  at  $p = 2$ ).

(i) For  $a_p = -2$  ( $\alpha = -1 + i$ ,  $\bar{\alpha} = -1 - i$ ):

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & 1/2 \\ -1 & 1/2 \end{pmatrix}.$$

(ii) For  $a_p = 0$  ( $\alpha = i\sqrt{2}$ ,  $\bar{\alpha} = -i\sqrt{2}$ ):

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & 0 \\ 0 & -1/2 \end{pmatrix} = -\frac{1}{2} \cdot I.$$

(iii) For  $a_p = +2$  ( $\alpha = 1 + i$ ,  $\bar{\alpha} = 1 - i$ ):

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & -1/2 \\ 1 & 1/2 \end{pmatrix}.$$

*Proof.* Direct matrix computation from  $A = \begin{pmatrix} a_p & 1 \\ -2 & 0 \end{pmatrix}$ .  $\square$

**3.2. Sprung half-logarithm constants are rational.** Combining Theorem 3.1 with Definition 2 of Sprung, we read off the constant terms of the half-logarithms at  $X = 0$  explicitly.

**Corollary 3.5** (Rationality of Sprung half-logarithm constants at  $p = 2$ ). *Let  $E$  be an elliptic curve supersingular at  $p = 2$ , with  $a_p \in \{-2, 0, +2\}$ . In the canonical  $(1, \alpha)$ -basis of  $\mathbb{Q}_2(\alpha)$ :*

(i) For  $a_p = -2$  ( $\alpha = -1 + i$ ):

$$(8) \quad \log_{\alpha}^{\#} |_{X=0} = \frac{1}{4}\alpha \in \mathbb{Q}\alpha,$$

$$(9) \quad \log_{\alpha}^b |_{X=0} = -\frac{1}{2} + \frac{1}{4}\alpha \in \mathbb{Q} + \mathbb{Q}\alpha.$$

(ii) For  $a_p = 0$  ( $\alpha = i\sqrt{2}$ ):

$$(10) \quad \log_{\alpha}^{\#} |_{X=0} = -\frac{1}{2} \in \mathbb{Q},$$

$$(11) \quad \log_{\alpha}^b |_{X=0} = -\frac{1}{4}\alpha \in \mathbb{Q}\alpha.$$

(iii) For  $a_p = +2$  ( $\alpha = 1 + i$ ):

$$(12) \quad \log_{\alpha}^{\#} |_{X=0} = -\frac{1}{4}\alpha \in \mathbb{Q}\alpha,$$

$$(13) \quad \log_{\alpha}^b |_{X=0} = \frac{1}{2} + \frac{1}{4}\alpha \in \mathbb{Q} + \mathbb{Q}\alpha.$$

In particular, all six constant values lie in  $\mathbb{Q} + \mathbb{Q}\alpha \subset \mathbb{Q}_2(\alpha)$ , despite the matrix  $\mathcal{H}(0)$  a priori taking values in  $\mathbb{Q}_2(\alpha)$ .

*Proof.* By Definition 2, the first column of the half-log matrix is

$$\begin{pmatrix} \log_{\alpha}^{\#} |_{X=0} \\ \log_{\alpha}^b |_{X=0} \end{pmatrix} = \mathcal{H}(0) \cdot \begin{pmatrix} 1 \\ -\alpha^{-1} \end{pmatrix}.$$

By Theorem 3.1,  $\mathcal{H}(0) = A^{-2}$ .

*Case  $a_p = -2$ .* Here  $\alpha = -1 + i$ , so  $\alpha^{-1} = -\frac{1}{2} - \frac{i}{2}$  and  $-\alpha^{-1} = \frac{1}{2} + \frac{i}{2}$ . Substituting the explicit  $A^{-2}$  from Proposition 3.4(i):

$$A^{-2} \cdot \begin{pmatrix} 1 \\ 1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -1/2 & 1/2 \\ -1 & 1/2 \end{pmatrix} \begin{pmatrix} 1 \\ 1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} + \frac{i}{4} \\ -\frac{3}{4} + \frac{i}{4} \end{pmatrix}.$$

We convert to the  $(1, \alpha)$ -basis using  $i = \alpha + 1$ :

$$\begin{aligned} \log_{\alpha}^{\#} |_{X=0} &= -\frac{1}{4} + \frac{i}{4} = -\frac{1}{4} + \frac{\alpha+1}{4} = \frac{1}{4}\alpha, \\ \log_{\alpha}^b |_{X=0} &= -\frac{3}{4} + \frac{i}{4} = -\frac{3}{4} + \frac{\alpha+1}{4} = -\frac{1}{2} + \frac{1}{4}\alpha. \end{aligned}$$

*Case  $a_p = 0$ .* Here  $\alpha = i\sqrt{2}$ , so  $\alpha^{-1} = -\frac{i\sqrt{2}}{2}$  and  $-\alpha^{-1} = \frac{i\sqrt{2}}{2}$ . Substituting  $A^{-2} = -\frac{1}{2} \cdot I$  from Proposition 3.4(ii):

$$A^{-2} \cdot \begin{pmatrix} 1 \\ i\sqrt{2}/2 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 1 \\ i\sqrt{2}/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{i\sqrt{2}}{4} \end{pmatrix}.$$

We convert using  $i\sqrt{2} = \alpha$ :

$$\log_{\alpha}^{\#} |_{X=0} = -\frac{1}{2}, \quad \log_{\alpha}^b |_{X=0} = -\frac{i\sqrt{2}}{4} = -\frac{1}{4}\alpha.$$

Case  $a_p = +2$ . Here  $\alpha = 1 + i$ , so  $\alpha^{-1} = \frac{1}{2} - \frac{i}{2}$  and  $-\alpha^{-1} = -\frac{1}{2} + \frac{i}{2}$ . The matrix  $A = \begin{pmatrix} 2 & 1 \\ -2 & 0 \end{pmatrix}$  has  $\det A = 2$  and  $A^{-2} = \begin{pmatrix} -1/2 & -1/2 \\ 1 & 1/2 \end{pmatrix}$ . Hence

$$A^{-2} \cdot \begin{pmatrix} 1 \\ -1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} - \frac{i}{4} \\ \frac{3}{4} + \frac{i}{4} \end{pmatrix}.$$

Using  $i = \alpha - 1$ :

$$\begin{aligned} \log_{\alpha}^{\sharp} |_{X=0} &= -\frac{1}{4} - \frac{i}{4} = -\frac{1}{4} - \frac{\alpha-1}{4} = -\frac{1}{4}\alpha, \\ \log_{\alpha}^b |_{X=0} &= \frac{3}{4} + \frac{i}{4} = \frac{3}{4} + \frac{\alpha-1}{4} = \frac{1}{2} + \frac{1}{4}\alpha. \end{aligned}$$

In all six cases the value lies in  $\mathbb{Q} + \mathbb{Q}\alpha$ , as claimed.  $\square$

**3.3. The SageMath–Sprung bridge.** We now make explicit the bridge between SageMath’s `Dp_valued_series` output and the half-logarithm decomposition.

**Theorem 3.6** (SageMath–Sprung bridge). *Let  $E/\mathbb{Q}$  be supersingular at  $p$ , and let  $\text{sp} = (v_0(T), v_1(T))$  denote the output of `Lp.Dp_valued_series(n, prec)`, where `Lp = E.padic_lseries(p)`. The output satisfies*

$$(14) \quad (1 - \varphi)^{-2} \cdot L_p(E, T) = v_0(T) \cdot \omega + v_1(T) \cdot \varphi(\omega)$$

in the Dieudonné basis  $(\omega, \varphi\omega)$ , where  $\varphi$  is the Frobenius matrix (5) on  $D_p(E)$ .

Equivalently, the  $(1, \alpha)$ -decomposition  $L_p(E, \alpha, T) = G(T) + H(T)\alpha$  is recovered from  $\text{sp}$  by the formula

$$(15) \quad (G_{\text{tmp}}, H_{\text{tmp}}) := \text{sp} \cdot ((1 - \varphi)^2)^T, \quad H = -\frac{H_{\text{tmp}}}{p}, \quad G = G_{\text{tmp}} - a_p H.$$

*Proof.* The first equation is the docstring statement of `Dp_valued_series(padic_lseries.py, class pAdicLseriesSupersingular, lines 1398–1447)`.

For the second equation, we first establish the algebraic identification between  $L_p(E, \alpha, T) = G(T) + H(T)\alpha \in \mathbb{Q}_p(\alpha)[[T]]$  and an element of  $D_p(E)[[T]]$  in the cyclic basis  $(\omega, \varphi\omega)$  of Section 2.3. There is a unique embedding  $\mathbb{Q}_p(\alpha) \hookrightarrow D_p(E)[1/p] \otimes \mathbb{Q}_p(\alpha)$  sending  $1 \mapsto \omega$  and respecting the multiplication  $\alpha^2 = a_p\alpha - p$  (the minimal polynomial of  $\alpha$  on the Tate module). Concretely, this embedding sends

$$(16) \quad \alpha \longmapsto a_p\omega - p\varphi(\omega).$$

To verify (16), compute the image of  $\alpha^2$  on both sides: on the source side,  $\alpha^2 = a_p\alpha - p$  maps to  $a_p(a_p\omega - p\varphi(\omega)) - p\omega = (a_p^2 - p)\omega - a_pp\varphi(\omega)$ ; on the target side,  $(a_p\omega - p\varphi(\omega)) \cdot (a_p\omega - p\varphi(\omega))$  in the ring  $\mathbb{Q}_p[\varphi]/(\varphi^2 - (a_p/p)\varphi + 1/p)$  equals  $a_p^2\omega - 2a_pp\varphi(\omega) + p^2\varphi^2(\omega) = a_p^2\omega - 2a_pp\varphi(\omega) + p^2((a_p/p)\varphi(\omega) - (1/p)\omega) = (a_p^2 - p)\omega - a_pp\varphi(\omega)$ , matching the source.

Under (16), the element  $L_p = G + H\alpha$  maps to

$$G \cdot \omega + H \cdot (a_p\omega - p\varphi(\omega)) = (G + a_pH)\omega + (-pH)\varphi(\omega).$$

Multiplying (14) from the right by  $(1 - \varphi)^2$  removes the normalisation factor, yielding  $L_p = (G + a_pH)\omega + (-pH)\varphi(\omega)$  in the cyclic basis. Solving for  $G$  and  $H$  from the components of  $\text{sp} \cdot ((1 - \varphi)^2)^T$  gives (15). This is exactly the inverse of the assignment in `padic_lseries.py`, lines 1442–1445.  $\square$

**Remark 3.7.** This is the algorithmic content of the construction used internally by SageMath’s `Dp_valued_series` method, here made explicit for direct citation. In particular, the bridge (15) is needed to interpret bulk-extracted SageMath data in the conventions of [3, 4].

#### 4. NUMERICAL VERIFICATION

We verified the SageMath–Sprung bridge of Theorem 3.6 on 180 elliptic curves drawn from the Cremona database [8], partitioned as follows: 75 curves with  $a_p(E) = -2$  in the conductor range [37, 997], and 105 curves with  $a_p(E) = 0$  in the conductor range [77, 63 096]. The  $a_p = 0$  block is subdivided into 75 low-conductor curves in [77, 225] and 30 curves in [1 001, 63 096] obtained via logarithmically-spaced bucket sampling. All curves have  $\text{rank}(E/\mathbb{Q}) \geq 1$ .

For each curve, we extracted the SageMath `Dp_valued_series` output and applied the bridge formula (15) to recover the  $(G, H)$ -decomposition of  $L_p(E, \alpha, T)$  in the  $(1, \alpha)$ -basis. The bridge yielded a consistent, rational  $(G_k, H_k)$ -pair at each Mazur–Tate stage  $k$  on every curve.

**4.1. Sample of verified curves.** Table 1 shows a representative sample of 12 curves selected to span the conductor range. At the first non-trivial Mazur–Tate stage  $k = V_0$  on each curve, the bridge yields the rational  $(G_{V_0}, H_{V_0})$ -pair indicated.

| Label   | Conductor | $a_p$ | $V_0$ | $(G_{V_0}, H_{V_0})$ |
|---------|-----------|-------|-------|----------------------|
| 17963c1 | 17 963    | −2    | 3     | (−2, −7/4)           |
| 70449a1 | 70 449    | −2    | 4     | (−2, −7/4)           |
| 53461b1 | 53 461    | −2    | 5     | (−1/4, −1)           |
| 55935g1 | 55 935    | −2    | 5     | (−2, −7/4)           |
| 30487a1 | 30 487    | 0     | 7     | (1, −1/4)            |
| 60803a1 | 60 803    | 0     | 7     | (3/2, −3/8)          |
| 58939a1 | 58 939    | 0     | 3     | (1, −1/4)            |

TABLE 1.  $(G_{V_0}, H_{V_0})$  for seven supersingular curves at  $p = 2$ , at the first stage  $V_0$  where  $L_p^b$  does not vanish (cf. Remark 4.1). Values are recovered from SageMath’s `Dp_valued_series` via the bridge formula (15); all seven pairs lie in  $\mathbb{Q}^2$  (a consequence of Theorem 3.6 and the rationality of SageMath’s modular-symbol output). The sample was chosen to span the full conductor range of the verified set; the remaining 173 curves yield analogous rational  $(G_{V_0}, H_{V_0})$ -pairs and are tabulated in the supplementary code repository. A reproducer script is provided at `code/sage_verify_paper_table1.py`.

**Remark 4.1** (Definition of  $V_0$  and rationality). We use  $V_0 = V_0^b$  to denote the first stage  $k \geq 0$  at which  $L_p^b$  is nonzero; equivalently, the smallest index  $k$  for which the second component  $v_1[k]$  of the SageMath output is nonzero. The Mazur–Tate stage  $V_0$  thus depends on the curve and is in general distinct

from  $X = 0$  (which is treated in Theorem 3.1); the table entries below are values of  $L_p$  at  $T^{V_0}$ , not at  $T = 0$ . The rational form of each entry follows from Theorem 3.6, since SageMath’s modular-symbol output is rational and the bridge formula (15) preserves rationality.

**Remark 4.2** (Asymmetric vanishing of  $L_p^\sharp$  and  $L_p^b$ ). In general,  $L_p^\sharp$  may vanish to a different order than  $L_p^b$ ; let  $V_0^\sharp$  and  $V_0^b = V_0$  denote the respective vanishing levels. The curve 55935g1 provides a concrete instance with  $V_0^\sharp = 4$  while  $V_0^b = 5$ : SageMath’s  $v_0[4] = 1/2 \neq 0$  at one stage before  $v_1$  becomes nonzero. The asymmetric vanishing phenomenon is intrinsic to Sprung’s framework and is discussed in [3, Section 6]. For all seven curves in Table 1, the  $(G_{V_0}, H_{V_0})$ -pair is recovered at the  $L_p^b$ -vanishing stage  $V_0^b$ .

**4.2. Computational details and code.** All verifications were conducted in SageMath 10 using the full Cremona database (largest conductor 499 998). The bridge formula (15) was implemented as a postprocessing step on the `Dp_valued_series` output, with no Frobenius-precision issues arising in the tested range. The verification code and the complete log files are publicly available at

<https://github.com/relkuch-del/sprung-h-matrix-p2>.

| Class                          | $n$        | Conductor range     | Bit-exact      |
|--------------------------------|------------|---------------------|----------------|
| $a_p = -2$ supersingular       | 75         | [37, 997]           | 75/75          |
| $a_p = 0$ supersingular (low)  | 75         | [77, 225]           | 75/75          |
| $a_p = 0$ supersingular (high) | 30         | [1 001, 63 096]     | 30/30          |
| <b>Total</b>                   | <b>180</b> | <b>[37, 63 096]</b> | <b>180/180</b> |

TABLE 2. Numerical verification of Theorem 3.6: for each curve, the inverse bridge formula (15) yields a rational  $(G_k, H_k)$  at every Mazur–Tate stage  $k$ , in bit-exact agreement with SageMath’s `Dp_valued_series` output. Verified on 180 curves spanning four orders of magnitude in conductor; zero discrepancies.

### 4.3. Summary.

**Remark 4.3** (The case  $a_p = +2$ ). The case  $a_p = +2$  at  $p = 2$  is covered theoretically by Proposition 3.4(iii) and Corollary 3.5(iii), but it does not occur for any of the 180 tested Cremona curves with  $\text{rank}(\mathbf{E}/\mathbb{Q}) \geq 1$ . We have  $\det((1 - \varphi)^2) = 1/4$  in this case.

## 5. DISCUSSION AND OPEN QUESTIONS

**5.1. Relation to Sprung’s main theorem at odd primes.** Sprung’s main theorem [4] concerns the case of an odd supersingular prime. The case  $p = 2$  remains open in the published literature. The explicit identification of Corollary 3.5 and the SageMath bridge of Theorem 3.6 provide concrete computational tools for further study of the  $p = 2$  case.



**5.2. Convergence of  $\mathcal{H}(X)$  for  $X \neq 0$ .** The analysis here treats only  $X = 0$ , where the limit (3) degenerates algebraically by Theorem 3.1. For  $X \neq 0$  the cofactor matrices  $C_k(X)$  depend nontrivially on  $k$ , and explicit power-series expansions of  $\mathcal{H}(X)$  in  $X$  require the convergence analysis of [3, §4]. We leave the explicit computation of higher-order terms to future work.

**5.3. Higher Mazur–Tate stages.** Theorem 3.6 translates SageMath output to the  $(1, \alpha)$ -basis at every Mazur–Tate stage  $k$ . The resulting sequences  $(G_k, H_k)_{k \geq 0}$  display rich rational structure across families of curves, including linearity patterns and stratifications by 2-adic valuation. We do not address these phenomena here; an empirical study of the higher-stage structure is the subject of separate work.

## REFERENCES

- [1] S. Kobayashi, *Iwasawa theory for elliptic curves at supersingular primes*, Invent. Math. **152** (2003), no. 1, 1–36. DOI: [10.1007/s00222-002-0265-4](https://doi.org/10.1007/s00222-002-0265-4).
- [2] R. Pollack, *On the  $p$ -adic  $L$ -function of a modular form at a supersingular prime*, Duke Math. J. **118** (2003), no. 3, 523–558. DOI: [10.1215/S0012-7094-03-11835-9](https://doi.org/10.1215/S0012-7094-03-11835-9).
- [3] F. E. I. Sprung, *Iwasawa theory for elliptic curves at supersingular primes: A pair of main conjectures*, J. Number Theory **132** (2012), no. 7, 1483–1506. DOI: [10.1016/j.jnt.2011.11.003](https://doi.org/10.1016/j.jnt.2011.11.003). arXiv: [0903.3419](https://arxiv.org/abs/0903.3419).
- [4] F. E. I. Sprung, *The Iwasawa Main Conjecture for elliptic curves at odd supersingular primes*, preprint, 2016. arXiv: [1610.10017](https://arxiv.org/abs/1610.10017).
- [5] F. Castella and X. Wan, *Iwasawa Main Conjecture for Heegner Points: Supersingular Case*, preprint, 2015/2016. arXiv: [1506.02538](https://arxiv.org/abs/1506.02538).
- [6] A. Lei, *Factorisation of two-variable  $p$ -adic  $L$ -functions*, Canad. Math. Bull. **57** (2014), no. 4, 845–852. arXiv: [1304.7473](https://arxiv.org/abs/1304.7473).
- [7] The Sage Developers, *SageMath, the Sage Mathematics Software System (Version 10.x)*, <https://www.sagemath.org>, 2026.
- [8] J. E. Cremona, *The Elliptic Curve Database for conductors up to 500 000*, <https://github.com/JohnCremona/ecdata>, accessed May 2026.

Email address: [relkuch@gmail.com](mailto:relkuch@gmail.com)